

Vidush Singhal

/vi.dʊʃ 'siŋ.gʱəl/

☎ +1-862-324-1823 | ✉ singhav@purdue.edu | 🏠 vidsinghal.github.io

📄 [vidush-singhal](#) | 🐙 [vidsinghal](#) | 🎓 [Google Scholar](#) |

707 N 5th St, Lafayette, IN, 47901

EDUCATION

- **Purdue University**, West Lafayette, IN, USA
Ph.D. in Electrical and Computer Engineering May 2021 – May 2027 (Expected)
- **Purdue University**, West Lafayette, IN, USA
M.S. in Electrical and Computer Engineering May 2021 – May 2023
- **Purdue University**, West Lafayette, IN, USA
B.Sc. in Computer Engineering Aug 2017 – May 2021

RESEARCH INTERESTS

I am a Ph.D. student in Electrical and Computer Engineering at Purdue University working on compiler techniques for improving the performance of programs over irregular recursive datatypes. My thesis is *Compiler Techniques for Optimizing the Layout of Irregular Recursive Datatypes*. My research interests include compilers, programming languages, program analysis, static analysis, data layout optimization, locality optimization, SIMD vectorization, automatic parallelization, and compilers for GPUs. I have experience building compiler infrastructure in Gibbon, LLVM, Clang, and MLIR/Torch-MLIR.

EXPERIENCE

- **Microsoft Research** [🌐] May 2025 – August 2025
Research Intern Redmond, WA, USA
 - Developed **C2Pulse**, an LLVM/Clang-based transpiler for verifying memory correctness of annotated C programs.
 - Implemented a **Clang-to-Clang** source transformer (~7K LOC of C++) that emits Pulse code for F* verification.
 - Built a C benchmark suite and validated generated verification conditions on representative annotated C programs.
 - Implemented and verified components of **DPE (Dice Protection Environment)** written in C through C2Pulse.
- **Lawrence Livermore National Laboratory** [🌐] May 2024 – August 2024
Computing Scholar Intern Livermore, CA, USA
 - Engineered an LLVM Attributor pass to **optimize byte layout within stack allocations** for improved spatial locality.
 - Implemented **LLVM IR** analyses and transformations to eliminate dead bytes and reorder frequently accessed bytes.
 - Optimized the **LLVM GPU sanitizer** by hoisting redundant checks out of loops, reducing XSBench runtime by **6%**.
 - Built an LLVM IR basic-block permutation pass to evaluate performance sensitivity to code layout.
- **Nod.ai Labs** [🌐] May 2022 – August 2022
Machine Learning Compiler Engineering Intern Santa Clara, CA, USA
 - Contributed to **Torch-MLIR**, a compiler stack for lowering programs from the **PyTorch ecosystem to MLIR**.
 - Implemented kernels and compiler support for multiple **PyTorch operators** in the Torch-MLIR pipeline.
 - Added support for the **Facebook DLRM** recommendation model in the PyTorch-to-MLIR compilation pipeline.
 - Investigated improvements to the **ILP solver** used by the Alpa scheduling framework.
- **Astrome Technologies Private Limited** [🌐] May 2019 – August 2019
Engineering Intern Bengaluru, Karnataka, India
 - Integrated an eMMC IP block with an AXI-Lite-to-Wishbone bridge using a SystemVerilog wrapper.
 - Verified the eMMC IP and bridge integration through Vivado behavioral simulations.
 - Built XSLT and Python tooling to render XML data as internal HTML reports.
 - Implemented convolutional encoding and Viterbi decoding algorithms in C for error correction and detection.
 - Achieved an **order-of-magnitude speedup** over the original Python implementation.
- **Purdue University** [🏠] May 2021 – Present
Graduate Research Assistant West Lafayette, IN, USA
 - **Gibbon Compiler**
 - * Optimized layouts of recursive datatypes to improve spatial locality.
 - * Measured speedups of **1.14x to 54x** over the best prior work.
 - * Implemented an array-of-structs to structure-of-arrays-style transformation for packed algebraic datatypes.

- * Structure-of-arrays transformation enables selective traversal, our benchmarks saw speedups of **1.1x to 12x**.
- * Developing optimizations enabled by the SoA representation such as loopification and vectorization.
- **Orchard Compiler**
 - * Developed a compiler framework to automatically fuse and parallelize tree traversals.
 - * Measured speedups ranging from **1x to 5x** over baseline tree traversals.
- **Copse Compiler**
 - * Developed a framework for vectorized evaluation of fully homomorphic encryption (FHE) decision trees.
- **Cornucopia**
 - * Developed a fuzzing-based framework to generate a corpus of optimized binaries from source programs using llc optimization flags.
 - * Discovered bugs in LLVM compiler optimization passes.
 - * Exposed decompilation failures in angr, Ghidra, and related binary analysis tools on generated binaries.
 - * Embedded ground-truth metadata in generated binaries to evaluate binary analysis tools (BATs).
- **Senior Design Project, SoCET Team** [[U](#)]

July 2020 – May 2021
 West Lafayette, IN, USA

Senior Design Team Member; advised by Dr. Mark Johnson and Prof. Samuel Midkiff

 - Developed an LLVM-based compiler tool for proprietary RISC-V hardware.
 - Implemented multiple LLVM RISC-V backend Machine IR passes to identify sparse program regions.
 - Annotated sparse regions so hardware could detect them at runtime and perform runtime checks.

SELECTED PROJECTS

- **Global Tensor Layout Optimization for IREE** June 2026
 Researched on techniques for global tensor layout optimization in the IREE compiler pipeline.
- **OpenMP GPU Offloading for Parallel STL Algorithms** Fall 2024
 Built a proof-of-concept evaluation of OpenMP GPU offloading for C++17 STL-style algorithms.

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, T=THESIS

- [C.1] Raghav Malik, **Vidush Singhal**, Benjamin Gottfried, Milind Kulkarni "Vectorized Secure Evaluation of Decision Forests" in PLDI 2021 [[ACM DL](#)]
- [C.2] **Vidush Singhal**, Akul Abhilash Pillai, Charitha Saumya, Milind Kulkarni, Aravind Machiry "Cornucopia: A Framework for Feedback Guided Generation of Binaries" in ASE 2022 [[ACM DL](#)]
- [C.3] **Vidush Singhal**, Chaitanya Koparkar, Joseph Zullo, Artem Pelenitsyn, Michael Vollmer, Mike Rainey, Milind Kulkarni "Optimizing Layout of Recursive Datatypes with Marmoset" in ECOOP 2024 [[LIPICs](#)]
- [J.1] **Vidush Singhal**, Laith Sakka, Kirshanthan Sundararajah, Ryan Newton, Milind Kulkarni "Orchard: Heterogeneous Parallelism and Fine-grained Fusion for Complex Tree Traversals" in TACO 2024 [[ACM DL](#)]
- [C.4] Chaitanya S. Koparkar, **Vidush Singhal**, Aditya Gupta, Mike Rainey, Michael Vollmer, Artem Pelenitsyn, Sam Tobin-Hochstadt, Milind Kulkarni, Ryan R. Newton "Garbage Collection for Mostly Serialized Heaps" in ISMM 2024 [[ACM DL](#)]

COURSEWORK

Compilers	Compiler Code Generation and Optimization; Compilers for GPUs; Introduction to Compilers and Translation Engineering
Programming Languages	Reasoning About Programs; Programming Languages; Advanced C Programming; Object-Oriented Programming in C++
Algorithms & Theory	Computation Models and Methods; Applied Algorithms; Data Structures in C; Formal Languages, Computability, and Parallelization
Systems	Computer Architecture; Programming Parallel Machines; Microprocessors and Microcontrollers; Digital Design
Mathematics	Discrete Mathematics; Graph Theory; Category Theory; Random Variables
AI & Data Science	Introduction to Artificial Intelligence; Python for Data Science
Security & Quantum	Holistic Software Security; Introduction to Quantum Computing

SKILLS

Programming Languages	C; C++; Haskell; Python; Java; Bash; MATLAB; Assembly (ARM, x86); SystemVerilog
Compiler & Verification	LLVM; GCC; Coq; Dafny; Pulse; GDB
Parallel & HPC	CUDA; OpenMP; MPI; Pthreads; OpenCilk; Intel TBB; Intel Cilk Tools
Data & Machine Learning	PyTorch; TensorFlow; NumPy
Systems & Databases	Unix/Linux; Windows; PostgreSQL; AFL++; gem5
Web & Development Tools	HTML; CSS; Git; Visual Studio Code; Vim; CLion; Notepad++; Kate
Professional Skills	Communication; teamwork; inclusive collaboration

TEACHING EXPERIENCE

- **Teaching Assistant**, Prof. Milind Kulkarni, Intro to Data Science, Purdue University *Fall 2025*
- **Teaching Assistant**, Prof. Milind Kulkarni, Intro to Data Science, Purdue University *Spring 2023*
- **Undergraduate Teaching Assistant**, Prof. Cheng Kok Koh, ECE 368, Purdue University *Spring 2020*
- **Student Grader for Signals and Systems**, Prof. Chih-Chun Wang, ECE 301, Purdue University *Spring 2020*

TALKS GIVEN

- **MWPLS 2025**, [AoS To SoA Transformation of packed ADTs](#) *Fall 2025*
- **MWPLS 2024**, Optimizing Layout of Recursive Datatypes with Marmoset *Fall 2024*
- **ECOOP 2024**, Optimizing Layout of Recursive Datatypes with Marmoset *Fall 2024*
- **PurPL Seminar**, Optimizing Layout of Recursive Datatypes with marmoset *Spring 2024*
- **ASE 2022**, Cornucopia: A Framework for Feedback Guided Generation of Binaries *Fall 2022*

HONORS AND AWARDS

- **Received a travel grant**, LLVM developers meeting *October 2024*
- **Semester Honors**, Purdue University *Fall 2019 - Spring 2020*
- **Deans's List and Semester Honors**, Purdue University *Fall 2017 - Spring 2018*

LEADERSHIP EXPERIENCE

- **Seminar Co-Coordinator**, Purdue Programming Languages and Systems Research Group (PurPL) *Fall 2025 - present*

REVIEW EXPERIENCE

- **Member of the Artifact Evaluation Committee (AEC)**, [ICFP 2026](#) *Summer 2026*
- **Member of the Program Committee (PC Member)**, [C3PO 2026](#) *Spring 2026*
- **Member of the Artifact Evaluation Committee (AEC)**, [PLDI 2026](#) *Spring 2026*
- **Member of the Artifact Evaluation Committee (AEC)**, [SPLASH 2026](#) *Spring 2026*
- **Additional Reviewer**, [PPoPP 2026](#) *Fall 2025*
- **Member of the Reviews for the Journal of Open Source Software**, [JOSS](#) *Fall 2025*
- **Member of the Artifact Evaluation Committee (AEC)**, [POPL 2026](#) *Fall 2025*
- **Member of the Artifact Evaluation Committee (AEC)**, [CGO 2026](#) *Fall 2025*
- **Member of the Artifact Evaluation Committee (AEC)**, [ICFP 2025](#) *Summer 2025*
- **Member of the Artifact Evaluation Committee (AEC)**, [PLDI 2025](#) *Spring 2025*
- **Collaborative Reviewer with Advisor (Milind Kulkarni)**, [CGO 2025](#) *Spring 2025*
- **Member of the Artifact Evaluation Committee (AEC)**, [CGO 2025](#) *Fall 2024*
- **Member of the Artifact Evaluation Committee (AEC)**, [PPoPP 2024](#) *Spring 2024*
- **Member of the Artifact Evaluation Committee (AEC)**, [PPoPP 2023](#) *Spring 2023*

VOLUNTEER EXPERIENCE

- **Talk moderator**, LLVM developers meeting 2024 *Fall 2024*
- **Student Volunteer**, ECOOP/ISSTA 2024 *Spring 2024*
- **Student Volunteer**, SPLASH 2021 *Fall 2021*

PROFESSIONAL MEMBERSHIPS

- **Eta Kappa Nu**, Purdue Electrical and Computer Engineering Honors Society *Fall 2018 - Present*

MENTORING

- **Logan Anderson**, Phd @ Purdue *Fall 2025*
- **Aniruddh Srivastava**, Undergrad @ Purdue *Fall 2025*
- **Peter A Kaya Gretchikha**, Undergrad @ Purdue *Fall 2024 - Fall 2025*
- **Mikah Kainen**, Undergrad @ Purdue *Fall 2023*
- **Qingyuan (Peter) Li**, Undergrad @ Purdue, SURF summer research mentor *Spring 2023 - Fall 2023*